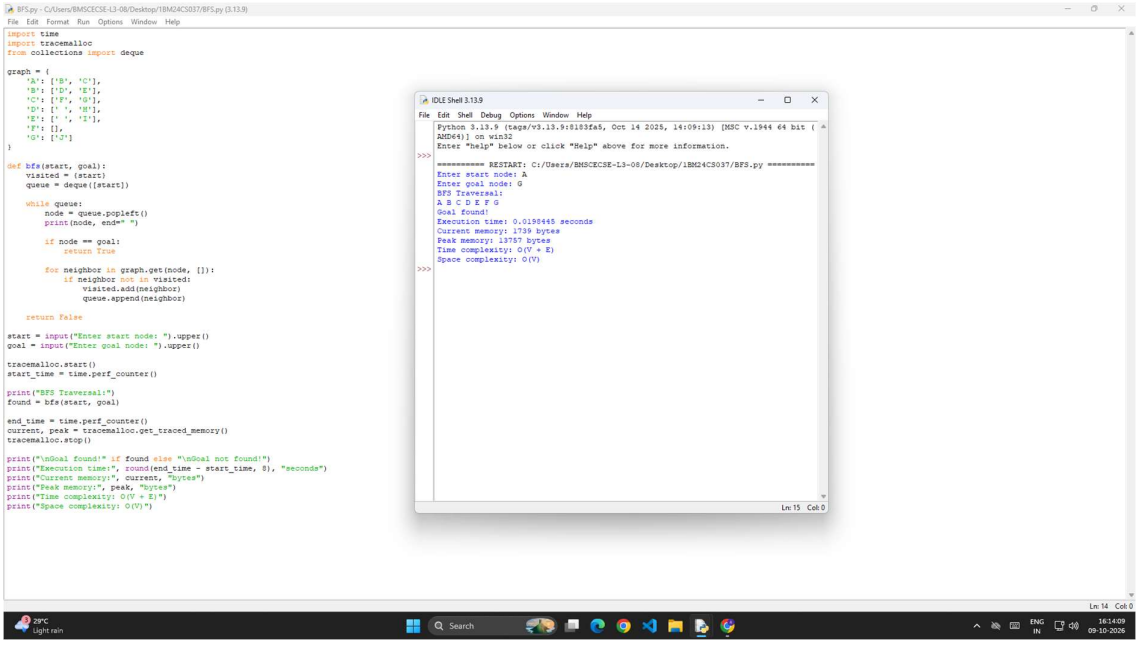
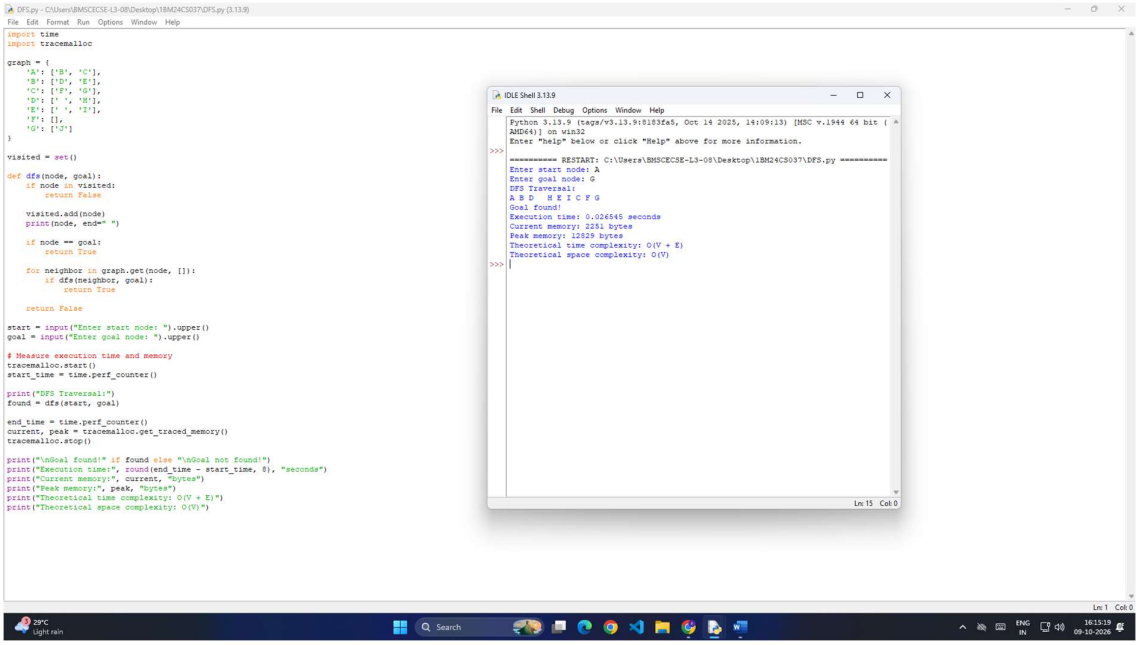
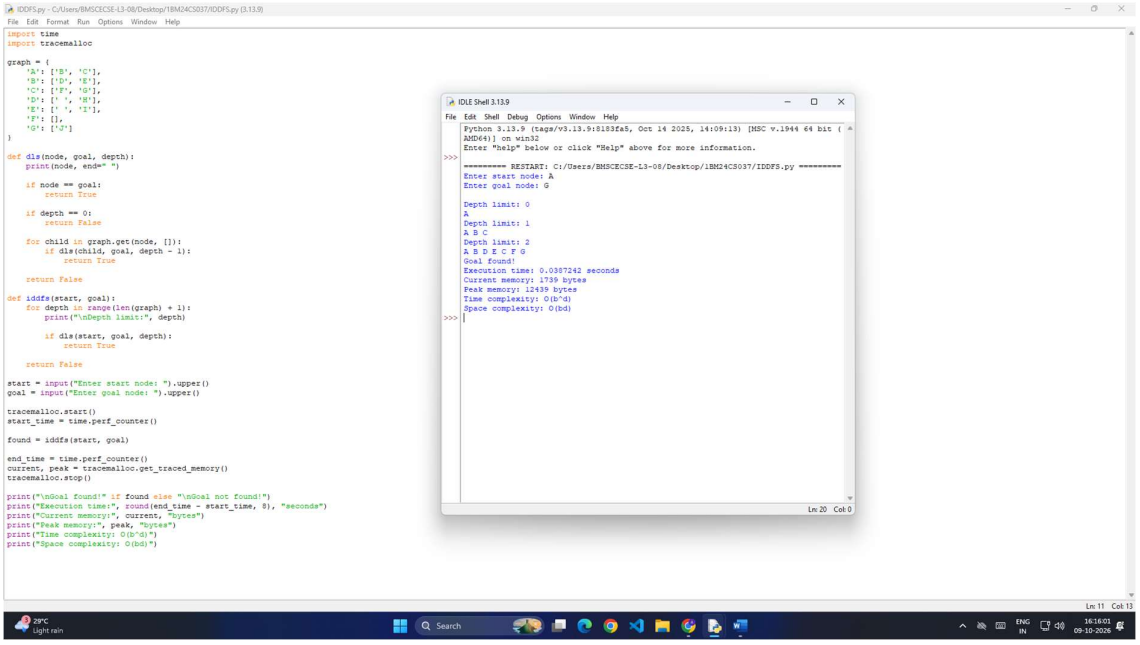
BFS


BFS.py - C:/Users/BMSCECSE-L3-08/Desktop/1BM24CS037/BFS.py (3.13.9)
File Edit Format Run Options Window Help
import time
import tracemalloc
from collections import deque
graph = {
'A': ['B', 'C'],
'B': ['D', 'E'],
'C': ['F', 'G'],
'D': [' ', 'H'],
'E': [' ', 'I'],
'F': [],
'G': ['J']
}
def bfs(start, goal):
visited = {start}
queue = deque([start])
while queue:
node = queue.popleft()
print(node, end=" ")
if node == goal:
return True
for neighbor in graph.get(node, []):
if neighbor not in visited:
visited.add(neighbor)
queue.append(neighbor)
return False
start = input("Enter start node: ").upper()
goal = input("Enter goal node: ").upper()
tracemalloc.start()
start_time = time.perf_counter()
print("BFS Traversal:")
found = bfs(start, goal)
end_time = time.perf_counter()
current, peak = tracemalloc.get_traced_memory()
tracemalloc.stop()
print("\nGoal found!" if found else "\nGoal not found!")
print("Execution time:", round(end_time - start_time, 8), "seconds")
print("Current memory:", current, "bytes")
print("Peak memory:", peak, "bytes")
print("Time complexity: O(V + E)")
print("Space complexity: O(V)")
Ln: 14 Col: 0
IDLE Shell 3.13.9
File Edit Shell Debug Options Window Help
Python 3.13.9 (tags/v3.13.9:8183fa5, Oct 14 2025, 14:09:13) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/BMSCECSE-L3-08/Desktop/1BM24CS037/BFS.py ==========
Enter start node: A
Enter goal node: G
BFS Traversal:
A B C D E F G
Goal found!
Execution time: 0.0198445 seconds
Current memory: 1739 bytes
Peak memory: 13757 bytes
Time complexity: O(V + E)
Space complexity: O(V)
>>>
Ln: 15 Col: 0
29°C
Light rain
Search
ENG
IN
16:14:09
09-10-2026

DFS


DFS.py - C:\Users\BMSCECSE-L3-08\Desktop\1BM24CS037\DFS.py (3.13.9)
File Edit Format Run Options Window Help
import time
import tracemalloc
graph = {
'A': ['B', 'C'],
'B': ['D', 'E'],
'C': ['F', 'G'],
'D': [' ', 'H'],
'E': [' ', 'I'],
'F': [],
'G': ['J']
}
visited = set()
def dfs(node, goal):
if node in visited:
return False
visited.add(node)
print(node, end=" ")
if node == goal:
return True
for neighbor in graph.get(node, []):
if dfs(neighbor, goal):
return True
return False
start = input("Enter start node: ").upper()
goal = input("Enter goal node: ").upper()
Measure execution time and memory
tracemalloc.start()
start_time = time.perf_counter()
print("DFS Traversal:")
found = dfs(start, goal)
end_time = time.perf_counter()
current, peak = tracemalloc.get_traced_memory()
tracemalloc.stop()
print("\nGoal found!" if found else "\nGoal not found!")
print("Execution time:", round(end_time - start_time, 8), "seconds")
print("Current memory:", current, "bytes")
print("Peak memory:", peak, "bytes")
print("Theoretical time complexity: O(V + E)")
print("Theoretical space complexity: O(V)")
Ln: 1 Col: 0
IDLE Shell 3.13.9
File Edit Shell Debug Options Window Help
Python 3.13.9 (tags/v3.13.9:8183fa5, Oct 14 2025, 14:09:13) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:\Users\BMSCECSE-L3-08\Desktop\1BM24CS037\DFS.py ==========
Enter start node: A
Enter goal node: G
DFS Traversal:
A B D H E I C F G
Goal found!
Execution time: 0.026545 seconds
Current memory: 2251 bytes
Peak memory: 12529 bytes
Theoretical time complexity: O(V + E)
Theoretical space complexity: O(V)
>>>
Ln: 15 Col: 0
29°C
Light rain
Search
ENG
IN
16:15:19
09-10-2026

IDDFS


IDDFS.py - C:/Users/BMSCECSE-L3-08/Desktop/1BM24CS037/IDDFS.py (3.13.9)
File Edit Format Run Options Window Help
import time
import tracemalloc
graph = {
'A': ['B', 'C'],
'B': ['D', 'E'],
'C': ['F', 'G'],
'D': [' ', 'H'],
'E': [' ', 'I'],
'F': [],
'G': ['J']
}
def dls(node, goal, depth):
print(node, end=" ")
if node == goal:
return True
if depth == 0:
return False
for child in graph.get(node, []):
if dls(child, goal, depth - 1):
return True
return False
def iddfs(start, goal):
for depth in range(len(graph) + 1):
print("\nDepth limit:", depth)
if dls(start, goal, depth):
return True
return False
start = input("Enter start node: ").upper()
goal = input("Enter goal node: ").upper()
tracemalloc.start()
start_time = time.perf_counter()
found = iddfs(start, goal)
end_time = time.perf_counter()
current, peak = tracemalloc.get_traced_memory()
tracemalloc.stop()
print("\nGoal found!" if found else "\nGoal not found!")
print("Execution time:", round(end_time - start_time, 8), "seconds")
print("Current memory:", current, "bytes")
print("Peak memory:", peak, "bytes")
print("Time complexity: O(b^d)")
print("Space complexity: O(bd)")
Ln: 11 Col: 13
IDLE Shell 3.13.9
File Edit Shell Debug Options Window Help
Python 3.13.9 (tags/v3.13.9:8183fa5, Oct 14 2025, 14:09:13) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========= RESTART: C:/Users/BMSCECSE-L3-08/Desktop/1BM24CS037/IDDFS.py =========
Enter start node: A
Enter goal node: G
Depth limit: 0
A
Depth limit: 1
A B C
Depth limit: 2
A B D E C F G
Goal found!
Execution time: 0.0387242 seconds
Current memory: 1739 bytes
Peak memory: 12439 bytes
Time complexity: O(b^d)
Space complexity: O(bd)
>>>
Ln: 20 Col: 0
29°C
Light rain
Search
ENG
IN
16:16:01
09-10-2026